

GENERATIVE AI FOR WORK & RESEARCH

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Title: Analysis of the relation between health and the economic level in the world

Table of content

I. Descriptive statistics and notable values	1
II. GDP per capita world map distribution	2
III. Life expectancy world map distribution	3
IV. Relationship between GDP and life expectancy.....	4
V. Add Your Reflection:	5

I. Descriptive statistics and notable values

Indicator	count	mean	std	min	25%	median	75%	max
GDP per capita	193	23,691	32,236	626	5,538	14,397	32,012	332,202
Life expectancy	193.0	72.0	7.2	53.8	66.6	72.6	77.1	85.7
Measure	Country			Value	Life expectancy		GDP per capita	
Lowest GDP/capita	South Sudan			626	55.8		NaN	
Highest GDP/capita	Monaco			332,202	79.6		NaN	
Lowest life expectancy	Central African Republic			53.8	NaN		1,137	
Highest life expectancy	Singapore			85.7	NaN		115,893	

Here is a clear interpretation of the summary statistics:

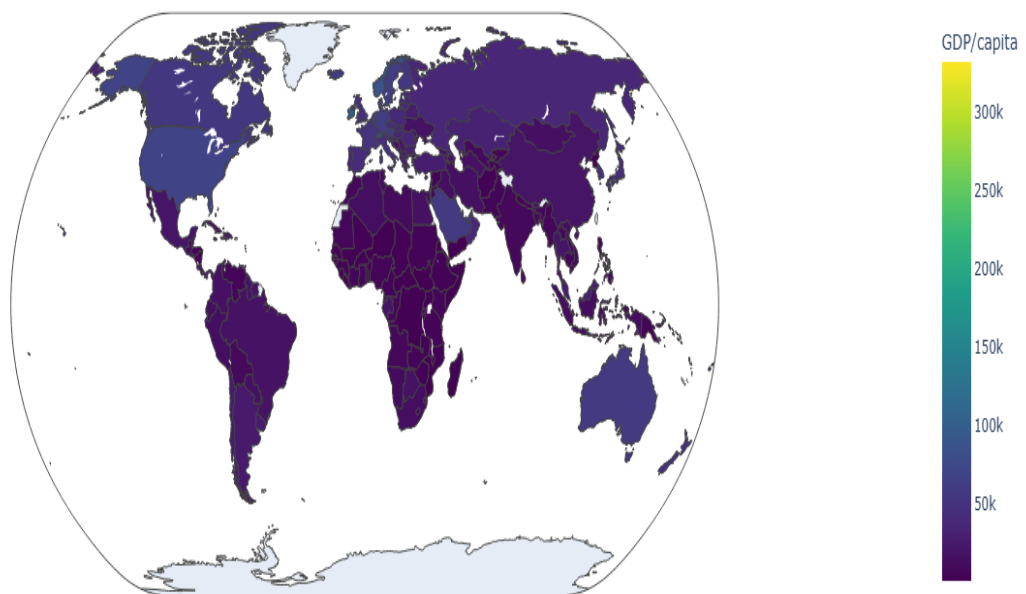
- The dataset includes 193 countries for both indicators in 2020. The average GDP per capita is about \$23,691, while the average life expectancy is about 72 years.
- GDP per capita shows a very large variation between countries. The lowest value is \$626 in South Sudan, while the highest is \$332,202 in Monaco. The

high standard deviation of \$32,236 shows that income levels are very unequal across countries.

- Life expectancy varies less than GDP per capita, but there are still important differences. The lowest life expectancy is 53.8 years in the Central African Republic, while the highest is 85.7 years in Singapore.
- The median GDP per capita is \$14,397, which is lower than the mean of \$23,691. This suggests that the GDP distribution is right-skewed, meaning a few very rich countries raise the average.
- The median life expectancy is 72.6 years, very close to the mean of 72.0 years, showing that life expectancy is more evenly distributed than GDP per capita.
- The extreme values suggest a strong global development gap: countries with very low GDP per capita often have lower life expectancy, while countries with higher GDP per capita generally tend to have longer life expectancy.

II. GDP per capita world map distribution

GDP per capita by country, 2020

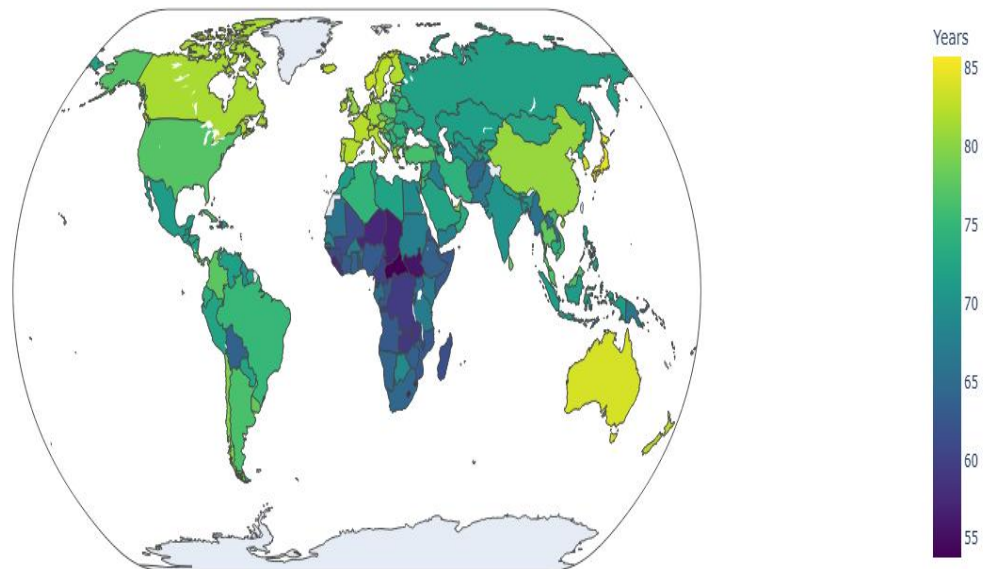


- The map shows GDP per capita by country in 2020, with darker purple representing lower GDP per capita and lighter green/yellow representing higher GDP per capita.
- High GDP per capita is concentrated in a few regions, especially North America, Western/Northern Europe, Australia, and some Gulf countries.
- Many countries in Africa, South Asia, and parts of Latin America appear in darker colors, indicating relatively lower GDP per capita in 2020.

- The map highlights a clear global economic inequality pattern, where wealth per person is unevenly distributed across countries, with a small number of countries having much higher GDP per capita than the global majority.

III. Life expectancy world map distribution

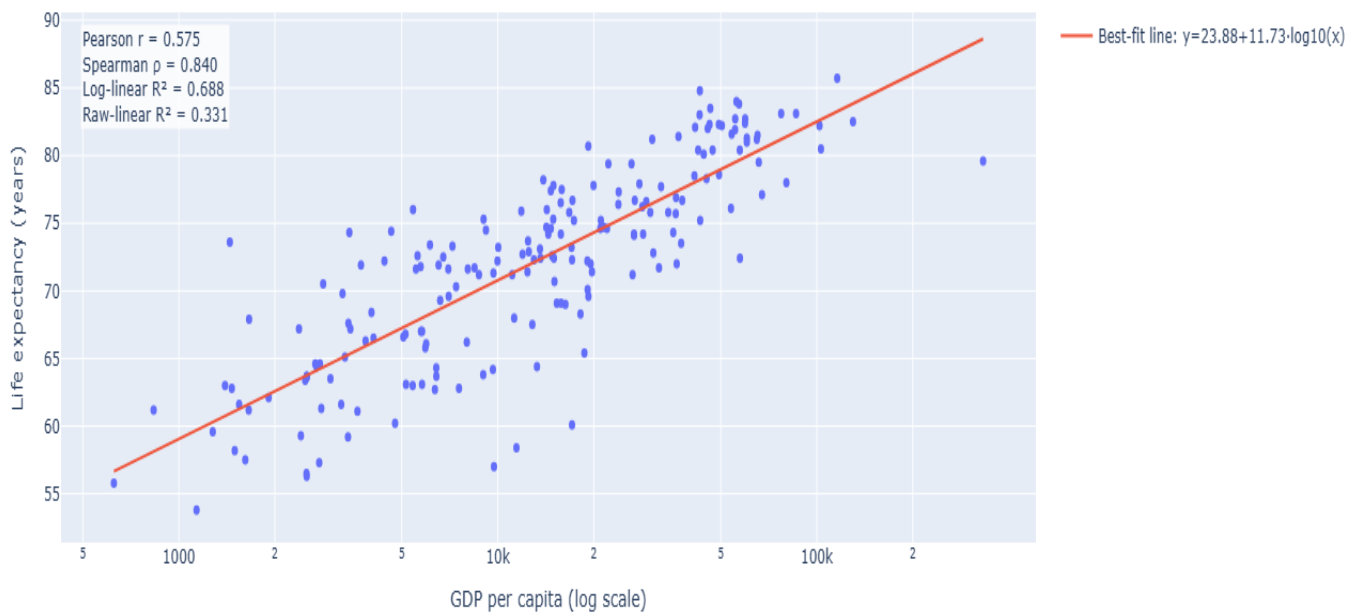
Life expectancy by country, 2020



- The map shows life expectancy by country in 2020, with darker colors representing lower life expectancy and lighter green/yellow colors representing higher life expectancy.
- Higher life expectancy is mostly found in wealthier regions, including Western Europe, North America, Australia, New Zealand, Japan, and parts of East Asia.
- Lower life expectancy is concentrated in parts of Sub-Saharan Africa, where several countries appear in darker blue and purple shades.
- Overall, the map shows a clear global inequality in health outcomes, with countries that likely have stronger healthcare systems, higher incomes, and better living conditions generally showing longer life expectancy.

IV. Relationship between GDP and life expectancy

Relationship between GDP per capita and life expectancy, 2020



Statistic	Value
Countries in merged 2020 data	193
Pearson correlation (raw GDP)	0.575
Spearman rank correlation	0.840
Raw linear regression	Life expectancy = $68.92 + 0.00012928 \times \text{GDP per capita}$
Raw linear R-squared	0.331
Log-linear regression	Life expectancy = $23.88 + 11.73 \times \log_{10}(\text{GDP per capita})$
Log-linear R-squared	0.688

There is a positive relationship between GDP per capita and life expectancy in 2020. The raw Pearson correlation is moderate ($r = 0.575$), while the Spearman rank correlation is strong ($\rho = 0.840$), showing that countries with higher GDP per capita usually have higher life expectancy. Because GDP per capita is highly skewed, the log-linear fitted line describes the scatter better: $R^2 = 0.688$, compared with $R^2 = 0.331$ for a raw linear model. This suggests diminishing returns: life expectancy rises rapidly as GDP increases among lower-income countries, then levels off among high-income countries.

V. Add Your Reflection

The process of working with the AI tool was very interesting. I realized that, to obtain useful results, it is important to know exactly what you want to achieve. Even more importantly, you need to translate your ideas and expectations into language that the AI tool can understand effectively.

This is not a process that can be completed perfectly in one attempt. Rather, it is an interactive process that requires refinement and clarification in order to achieve good results.

One of the main challenges was understanding the different options and capabilities of the AI tool. However, by the end of the process, I realized that the conversation with the tool is quite intuitive and that the AI is able to understand a wide range of instructions.

Finally, it is very important to carefully review the AI's output to make sure that the results match exactly what was requested.